
Chapter 3: Life Processes in Living Organisms – Part 2

Topic: Reproduction

Important Topics

- Asexual and Sexual reproduction
- Reproduction and modern technology
- Reproductive health
- Population Explosion

What are Life Processes?

Life processes are the basic activities carried out by living organisms to maintain life.

Important life processes:

- Nutrition
- Respiration
- Excretion
- Sensation and Response (Control & Coordination)
- **Reproduction**

👉 All life processes are essential for survival **except reproduction**.

Why is Reproduction Important?

- Reproduction does **not help an individual to survive**
- It helps in **maintenance and continuity of the species**
- It is important for **evolution of species**

Maintenance of Species

Maintenance of species means:

Continuation of a species by producing new individuals of the same kind.

Reproduction – Definition

Formation of a new organism of the same species by an existing organism is called reproduction.

Types of Reproduction

There are **two main types** of reproduction:

1. **Asexual reproduction**
 2. **Sexual reproduction**
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Asexual Reproduction

Definition:

Asexual reproduction is the process of formation of new organism without involvement of gametes.

Characteristics:

- Only **one parent** is involved (Uniparental)
- No formation or fusion of gametes
- New organism is **genetically identical** to parent
- Occurs by **mitotic cell division**
- No genetic recombination

Advantages:

- Fast process
- Large number of organisms produced quickly

Disadvantage:

- No variation → less adaptability

Asexual Reproduction in Unicellular Organisms

1. Binary Fission

Definition:

Binary fission is a type of asexual reproduction in which one parent cell divides into two identical daughter cells.

Organisms showing binary fission:

- Bacteria (Prokaryotes)
- Amoeba
- Paramecium
- Euglena
- Cell organelles like mitochondria and chloroplasts

Types of Binary Fission:

1. **Simple Binary Fission** – Amoeba
 - Division occurs in any plane
 - Due to irregular shape
2. **Transverse Binary Fission** – Paramecium
 - Division occurs across the body
3. **Longitudinal Binary Fission** – Euglena
 - Division occurs lengthwise

Important Points:

- Occurs in **favourable conditions**
- Division may be by **mitosis or amitosis**

2. Multiple Fission

Definition:

Multiple fission is a type of asexual reproduction in which the parent cell divides to form many daughter cells at the same time.

Example:

- Amoeba

Process:

- Occurs during **unfavourable conditions**
 - Amoeba forms a protective covering called **cyst**
 - Repeated nuclear divisions occur
 - Cytoplasm divides to form many **amoebulae**
 - When conditions become favourable, cyst breaks and amoebulae are released
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Key Terms (Very Important for Exam)

- **Cyst:** Protective covering formed during adverse conditions
 - **Amoebulae:** Small daughter Amoeba formed by multiple fission
 - **Mitotic division:** Cell division for growth and asexual reproduction
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Does the parent cell exist after asexual reproduction (fission)?

Answer:

✗ No, the parent cell does **not exist** after fission because it divides completely to form new daughter cells.

3. Budding (Asexual Reproduction)

Budding in Yeast

Definition:

Budding is a type of asexual reproduction in which a new organism develops as an outgrowth (bud) on the parent body.

Process:

- Yeast is a **unicellular fungus**

- Parent yeast cell divides its nucleus by **mitosis**
- A small bulge appears on parent cell called **bud**
- One daughter nucleus enters the bud
- Bud grows, separates and lives independently

Important Points:

- Parent cell exists after budding
 - Occurs in **favourable conditions**
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B. Asexual Reproduction in Multicellular Organisms

1. Fragmentation

Definition:

Fragmentation is a type of asexual reproduction in which the parent body breaks into fragments and each fragment develops into a new organism.

Examples:

- **Spirogyra** (algae)
- **Sycon** (sponge)

Explanation:

3. In Spirogyra, filaments break into fragments
 4. Each fragment grows into a new Spirogyra
 5. In Sycon, accidental body fragments grow into new individuals
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2. Regeneration

Definition:

Regeneration is the ability of an organism to regrow its lost body parts or form a complete organism from body fragments.

Examples:

- **Lizard** – tail regeneration (limited regeneration)
- **Planaria** – whole body regeneration

Important Point:

- In Planaria, body splits into two parts
 - Each part regenerates remaining body
 - Two new Planaria are formed
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3. Budding in Hydra

- Occurs in **favourable conditions**
 - A bud develops due to repeated division of regenerative cells
 - Bud remains attached and gets nutrition from parent
 - When fully grown, bud separates and lives independently
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4. Vegetative Propagation (Plants)

Definition:

Vegetative propagation is a type of asexual reproduction in plants using vegetative parts like root, stem, leaf or bud.

Examples:

- **Potato** – eyes on tuber
- **Bryophyllum** – buds on leaf margin
- **Sugarcane, grasses** – buds on nodes
- **Carrot, radish** – roots

Advantage:

4. Fast reproduction
5. New plants identical to parent

5. Spore Formation

Definition:

Spore formation is a type of asexual reproduction in which spores develop into new organisms under favourable conditions.

Example:

- Fungi (Mucor)

Process:

- Sporangia produce spores
- Sporangium bursts and releases spores
- Spores germinate in warm, moist conditions
- New fungal colony forms

Sexual Reproduction

Definition:

Sexual reproduction is a type of reproduction involving fusion of male and female gametes.

Main Processes in Sexual Reproduction

1. Gamete Formation

- Occurs by **meiosis**
 - Produces **haploid gametes**
 - Chromosome number is halved
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2. Fertilization

- Fusion of male and female gametes
- Forms **diploid zygote**
- Zygote divides by mitosis
- Embryo develops into new individual

Importance:

- Causes **genetic variation**
- Helps in adaptation and survival
- Prevents extinction

Sexual Reproduction in Plants

Flower – Reproductive Structure

Flower is the **structural unit** of sexual reproduction in plants.

Floral Whorls:

- **Calyx** – Sepals (green, protection)
- **Corolla** – Petals (coloured)
- **Androecium** – Male whorl (stamens)
- **Gynoecium** – Female whorl (carpels)

Essential Whorls:

- Androecium
- Gynoecium

Accessory Whorls:

- Calyx
 - Corolla
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Types of Flowers

- **Bisexual flower** – Both androecium & gynoecium
👉 Example: Hibiscus
 - **Unisexual flower** – Only one whorl
👉 Example: Papaya
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Structure of Gynoecium

- **Ovary** – Contains ovules
- **Style** – Connects ovary & stigma
- **Stigma** – Sticky, receives pollen

Each ovule contains:

- Embryo sac
 - Haploid egg cell
 - Two polar nuclei
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Pollination

Definition:

Transfer of pollen grains from anther to stigma is called pollination.

Types:

- **Self-pollination**
- **Cross-pollination**

Agents:

- Abiotic: Wind, water
 - Biotic: Insects, animals
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Fertilization in Angiosperms (Double Fertilization)

- Pollen grain germinates on stigma
 - Pollen tube carries two male gametes
 - One male gamete + egg → **Zygote**
 - Second male gamete + polar nuclei → **Endosperm**
 - Called **double fertilization**
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Seed Germination

Definition:

Development of a seed into a new plant under favourable conditions is called seed germination.

Process:

- After fertilization:
 - **Ovule → Seed**
 - **Ovary → Fruit**
 - Seeds fall on soil
 - Zygote uses food stored in **endosperm**
 - A new **plantlet** is formed
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Sexual Reproduction in Human Beings

Sex Chromosomes in Humans

- **Male:** XY
 - **Female:** XX
 - X-chromosome present in both
 - Y-chromosome present only in males
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Human Male Reproductive System

Main Parts:

- Testes
- Epididymis
- Vas deferens
- Ejaculatory duct
- Urinogenital duct
- Penis
- Glands:
 - Seminal vesicles
 - Prostate gland
 - Cowper's gland

Functions:

6. Testes:

1. Present in scrotum
2. Contain seminiferous tubules
3. Produce sperms by **meiosis**

7. Sperms pass through ducts:

1. Rete testes → Vas eferens → Epididymis → Vas deferens → Ejaculatory duct → Urinogenital duct

8. Sperms mature while passing through ducts

9. **Semen** = Sperms + gland secretions

10. Semen is ejaculated through penis

Human Female Reproductive System

Main Parts:

- Pair of ovaries
- Pair of oviducts (fallopian tubes)
- Uterus
- Vagina

Functions:

- Ovary releases **one ovum every month**
 - Oocyte enters oviduct through funnel-shaped end
 - Cilia in oviduct push ovum towards uterus
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Gamete Formation

- Both **sperms and ova** are formed by **meiosis**
 - Sperms:
 - Produced from puberty till death
 - Ova:
 - 2–4 million immature oocytes at birth
 - Only about **400 oocytes** released till menopause
 - **Menopause:**
 - Occurs at 45–50 years
 - Stoppage of menstrual cycle
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Fertilization in Humans

Definition:

Formation of zygote by fusion of sperm and ovum is called fertilization.

Important Points:

- Fertilization is **internal**
 - Occurs in **oviduct**
 - Millions of sperms released, but only **one fertilizes the ovum**
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Chromosome Number

6. Germ cells: **Diploid** ($2n = 44 + XX / XY$)

7. Gametes: **Haploid** ($n = 22 + X / Y$)

8. Sperms:

- Two types $\rightarrow (22 + X)$ or $(22 + Y)$

9. Ova:

- Only one type $\rightarrow (22 + X)$

Sex Determination:

- Male decides sex of child
 - X sperm $\rightarrow$ Girl (XX)
 - Y sperm $\rightarrow$ Boy (XY)
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Development and Birth

- Zygote divides by mitosis $\rightarrow$ Embryo
 - Embryo implants in uterus
 - **Placenta** supplies food and oxygen
 - Development completed in about **9 months**
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Menstrual Cycle

Definition:

Cyclic changes occurring in female reproductive system every 28–30 days are called menstrual cycle.

Hormones Involved:

- Follicle Stimulating Hormone (FSH)
- Luteinizing Hormone (LH)
- Estrogen
- Progesterone

Phases:

- **Menstrual phase**

- Regeneration phase
- Ovulation
- Secretory phase

Key Points:

- FSH → Growth of follicle
 - Estrogen → Development of endometrium
 - LH → Ovulation
 - Progesterone → Prepares uterus for implantation
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Social Message (Very Important)

- Mother is **not responsible** for girl child
 - Male determines sex of child
 - Female foeticide must be stopped ❌
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Menstruation

Definition:

The periodic discharge of blood, unfertilized ovum and degenerated endometrium through vagina is called menstruation.

Process:

- If ovum is **not fertilized within 24 hours**:
 - Corpus luteum becomes inactive
 - It converts into **corpus albicans**
- Secretion of **estrogen and progesterone stops**
- Endometrium degenerates
- Degenerated tissues and ovum are expelled
- Bleeding lasts for about **5 days**

Important Points:

- Occurs every month
- Stops temporarily during:
 - Pregnancy
 - Breastfeeding
- Menstruation is a **natural process**

Care During Menstruation:

- Proper rest is needed
 - Personal hygiene must be maintained
 - Risk of infection is high during this period
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Reproduction and Modern Technology

Some couples cannot have children due to problems like:

- Irregular menstrual cycle
- Problems in ovulation
- Blockage in oviduct
- Low sperm count
- Abnormal sperms

Modern techniques help such couples.

In Vitro Fertilization (IVF)

Definition:

Fertilization carried out outside the body in a test tube is called IVF.

Process:

- Fertilization occurs in laboratory
- Embryo is implanted in uterus

Used when:

- Low sperm count
 - Blocked oviduct
 - Implantation problems
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Surrogacy

Definition:

Implantation of embryo in the uterus of another woman is called surrogacy.

Process:

- Ovum collected from mother
 - Fertilized with husband's sperm in test tube
 - Embryo implanted in uterus of another woman
 - That woman is called **surrogate mother**
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Sperm Bank / Semen Bank

- Similar to blood bank
 - Semen collected from healthy men
 - Stored after medical check-up
 - Used for couples having male infertility
 - Identity of donor is kept **secret by law**
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Twins

Definition:

Two offsprings born at the same time are called twins.

Types of Twins:

1. Monozygotic Twins

- Formed from **single zygote**
- Embryo splits into two
- Genetically **identical**
- Same gender
- Same appearance

⚠ If splitting occurs late → **Conjoined (Siamese) twins**

2. Dizygotic Twins

- Two ova released
 - Fertilized by two different sperms
 - Two zygotes formed
 - Genetically **different**
 - Gender may be same or different
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Reproductive Health

Definition:

A state of physical, mental and social well-being related to reproductive system is called reproductive health.

Important Points:

- Awareness about reproductive health is low
 - Women need special care during menstruation
 - Cleanliness of genitals is important for both men and women
 - Lack of hygiene can cause infections
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Sexually Transmitted Diseases (STDs)

1. Syphilis

- Caused by bacteria
- Symptoms:
 - Chancre (patches)
 - Fever
 - Joint pain
 - Hair loss

2. Gonorrhea

- Caused by bacteria
 - Symptoms:
 - Burning urination
 - Pus discharge
 - Infection of urinary tract
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